Flash Storage and SSDs

Flash Storage

Flash storage is a type of non-volatile memory that retains data without power. It is mainly available in two forms: NOR Flash and NAND Flash.

NOR Flash

Characteristics of NOR Flash

- Provides direct byte-level random access.
- Faster read speeds for code execution (eXecute In Place – XIP).
- More expensive than NAND flash.
- Larger cell size $\rightarrow$ lower storage density.
- Typical use: embedded systems, BIOS, firmware.

NAND Flash

Characteristics of NAND Flash

- Widely used for data storage (e.g., SSDs, USB drives, memory cards).
- Cheaper and higher density than NOR flash.
- Requires **page-at-a-time read** (512 bytes to 4 KB).
- **Read latency:** 20–100 μs per page.
- Similar performance for sequential vs. random read.
- **Write constraint:** Page can only be written once, must be erased before rewrite.
- **Erase unit:** Erase block (typically 256 KB to 1 MB, i.e., 128–256 pages).
- **Erase latency:** 2–5 ms.

Solid-State Disks (SSDs)

SSDs Built on NAND Flash

- Use standard block-oriented disk interfaces.
- Internally store data on multiple NAND flash chips.
- Transfer rates: up to 500 MB/s (SATA), up to 3 GB/s (NVMe PCIe).
- Remapping of logical page $\leftrightarrow$ physical page addresses avoids waiting for erases.
- **Flash Translation Table (FTT):** Maintains mapping and is stored in flash label fields.
- Managed by **Flash Translation Layer (FTL)**.

Reliability and Wear Leveling

Flash Reliability

- Flash cells wear out after repeated erase cycles.
- **Endurance:**
  - SLC (Single-Level Cell): $\approx$ 100,000 erases.
  - MLC (Multi-Level Cell): $\approx$ 10,000 erases.
  - TLC (Triple-Level Cell): $\approx$ 1,000–3,000 erases.
  - QLC (Quad-Level Cell): $\approx$ 1,000 erases.
- **Wear leveling:** Cold (infrequently updated) data is stored in blocks already erased many times to balance wear.

Figure 1: Comparison of SLC, MLC, TLC, and QLC flash storage technologies.